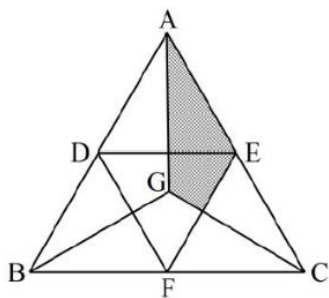




TOUR DE MATHS – MAY 2019

Level ONE (5 marks per question)

- 1** An equilateral triangle ABC is shown alongside. The centre G is joined to A, B and C forming three identical triangles. D, E and F are the midpoints of AB, AC and BC respectively.
- If the area of $\triangle ABC$ is 120 cm^2 , what is the shaded area in cm^2 ?



- 2** For how many positive integers n less than 50 is the product $(n - 8)(n - 38)$ positive?
- 3** In a pet shop, the ratio of hamsters to kittens is $3 : 5$, and the ratio of birds to kittens is $5 : 8$. What is the ratio of hamsters to birds?
- 4** The formula for converting $^{\circ}\text{F}$ into $^{\circ}\text{C}$ is $C = \frac{5}{9}(F - 32)$. The temperature in $^{\circ}\text{C}$ which is twice as big when converted to $^{\circ}\text{F}$ is...

- 5** The diagrams show three scales. On each scale there are different objects on each side which balance each other, as shown.

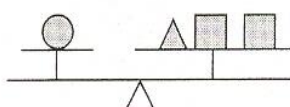


Diagram 1

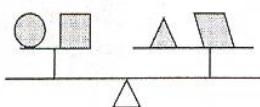


Diagram 2

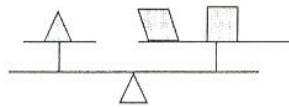
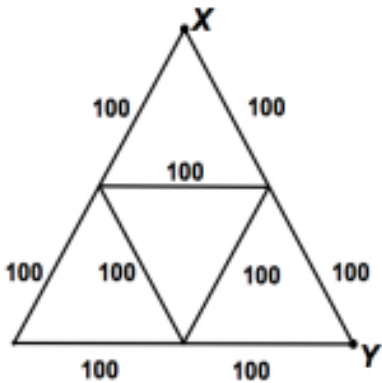
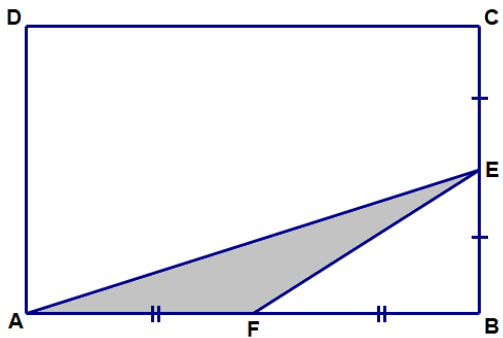
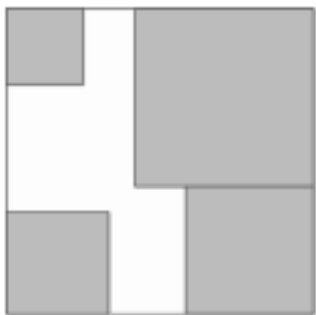


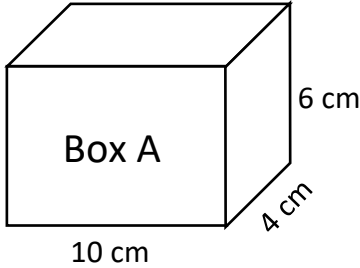
Diagram 3

How many 'square' shaped objects will balance a 'circle' object?

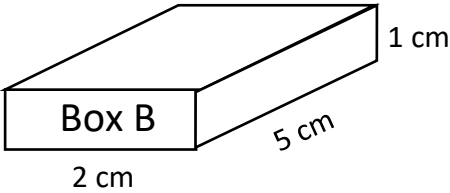
- 6** A frog is at the bottom of a well that is 30m deep. Each hour he climbs 3m and then slips back 2m. How many hours does he take to get out of the well?

7	The greatest number of Fridays that can occur in a 45-day period is...?																										
8	If $p \otimes q$ means $3p + q^2$ then $(3 \otimes 4) \otimes 5 = ?$																										
9	Each of the nine paths in a park is 100m long. Ann wants to go from X to Y without going along any path more than once. What is the length, in metres, of the longest route she can choose?	10	ABCD is a rectangle. F and E are midpoints of AB and CB respectively. If the area of the shaded $\triangle AEF$ is 7cm^2 , what is the area of ABCD in cm^2 ?																								
																											
11	In the four corners of the big square in the figure, squares with areas 9, 16, 25 and 49 are shaded. The two larger squares are touching. What is the area of the unshaded part of the big square?	12	In the grid below, the squares in each row and each column contain a number from 1 to 4, and each number is used only once in each row and column. Inequality signs indicated the relationship between certain squares. What number is in Square A?																								
		<table border="1" data-bbox="932 1227 1283 1599"> <tr> <td>A</td><td></td><td></td><td></td></tr> <tr> <td></td><td>^</td><td></td><td>v</td></tr> <tr> <td></td><td></td><td></td><td></td></tr> <tr> <td></td><td><</td><td>3</td><td>></td></tr> <tr> <td></td><td></td><td></td><td>v</td></tr> <tr> <td></td><td></td><td></td><td></td></tr> </table>		A					^		v						<	3	>				v				
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Level TWO (10 marks per question)																											
13	A three-digit number has $(2x-1)$ as its units digit and x as its hundreds digit. This number is represented by $112x+29$. Find, in terms of x , the tens digit.																										
14	At a school 15 students were absent on Monday, 12 were absent on Tuesday and 9 were absent on Wednesday. If 22 students were absent at least once over the 3 days, what is the maximum number of students who could have been absent on all three days?																										

15 How many boxes identical to Box B will fill Box A exactly?



Box A



Box B

16 Each letter in the addition sum shown below stands for a different digit, with *S* standing for 3.

$$\begin{array}{r} SO \\ + MANY \\ \hline SUMS \end{array}$$

What is the value of $Y \times O$?

17 Kyle takes 8 subjects at school, and for each one receives a mark from 1 to 7. If the sum of his marks is 50, then the least number of 7s he must have received is...

18 In a hockey tournament, each team played each other team once.

The final league table was:

	WINS	DRAWS	LOSSES	POINTS
BULLS	1	2	0	4
CHEETAHS	1	1	1	3
LIONS	1	1	1	3
SHARKS	1	0	2	2

If the Sharks beat the Cheetahs, then which of the following statements is true? Write the letter of the true statement.

A) Lions defeated Cheetahs, but lost to Sharks.
 B) Bulls won against either Cheetahs or Lions.
 C) In matches against Sharks, Cheetahs were more successful than Lions.
 D) In matches against Lions, Bulls were more successful than Cheetahs.
 E) Lions were undefeated, except against Cheetahs.

19 Luc takes a taxi to his home but falls asleep when the taxi is halfway to his house. He wakes up when the remaining part of his journey is equal to half the distance covered by the taxi while he was asleep.

If the fraction of the journey for which he slept is $\frac{1}{n}$, what is the value of n ?

20 A positive integer is to be placed in each box below. Integers may be repeated, but the product of any four adjacent integers is always 120.

Determine the sum of all possible values of x .

		2			4			x			3		
--	--	---	--	--	---	--	--	-----	--	--	---	--	--

Level THREE – 15 marks per question

- 21** At the start of the Mighty Mathematicians Football Team's first game of the season, their coach noticed that the jersey numbers of the 22 players on the field (11 players per team) were all the numbers from 1 to 22. At half-time, the coach substituted her goalkeeper (who had the number 1 on her jersey) for a reserve player. The coach then noticed that after the substitution, no two players on the field had the same jersey number and that the sum of the jersey numbers of each of the teams were exactly equal. What is the greatest (positive) possible jersey number of the reserve player?

- 22** Different letters are assigned different positive integer values and these values are then multiplied together to determine the value of certain words.

For example, if $C=4$, $A=8$ and $T=12$, then $CAT = 4 \times 8 \times 12 = 384$.

Given that $HILL = 15$

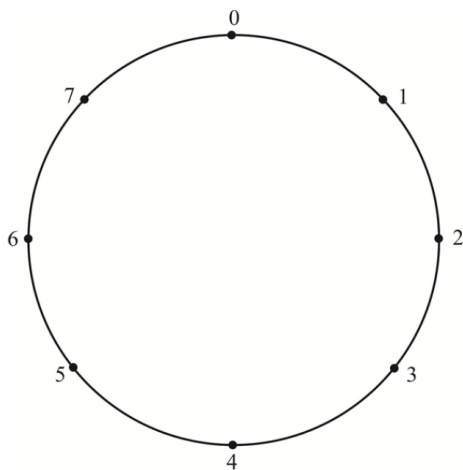
$PHOTO = 8470$

$HILLS = 195$

$HIPHOP = 3300$

Find the value of $PITSTOP$.

- 23** In modular arithmetic, numbers are arranged in a circle, similar to the face of a clock. Addition is represented by a clockwise movement; subtraction is represented by an anti-clockwise movement. The diagram below shows a modular 8 circle.



Examples: $2 + 4 = 6$

$3 + 6 = 1$

$1 - 4 = 5$

$3 \times 5 = 15 = 7$

According to the modular 8 circle given, what is the value of $4 \times 2 - (3 - 5)$?

- 24** Four people need to cross a rickety bridge at night. Unfortunately, they have only one torch and the bridge is too dangerous to cross without a torch. Furthermore, the bridge is only strong enough to support two people at a time.

Not all of the people take the same amount of time to cross the bridge either. The fastest time for each person are 1 min, 2 mins, 7 mins and 10 mins, respectively (they can take longer).

What is the shortest time needed for all four of them to cross the bridge?